

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 30 September 2026

Team ID: SWTID-2026-5226

Project Name: Study Buddy API

Maximum Marks: 4 Marks

Technical Architecture:

The Study Buddy API architecture supports student authentication, study task and timetable management, notes and resources, progress tracking, and optional AI-assisted study features.

Architecture Flow: Client / Web or Mobile UI → REST API → Authentication & Study Services → PostgreSQL Database / Cloud Storage → Optional Notification, Calendar and AI Services.

Guidelines:

- Include all the Study Buddy API processes as application logic / technology blocks.
- Provide infrastructural demarcation (Local / Cloud).
- Indicate external interfaces such as notification, calendar and AI APIs.
- Indicate data storage components/services for users, tasks, notes and resources.
- Indicate the interface to AI/ML models if AI-based study assistance is enabled.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	API Interface	REST API through which students access Study Buddy features.	Python / FastAPI
2	Application Logic-1	User registration, login and authentication.	Python / FastAPI
3	Application Logic-2	Study task, timetable and reminder management.	Python / FastAPI
4	Application Logic-3	Study notes, resources and progress management.	Python / FastAPI
5	Database	Stores users, subjects, tasks, notes and study progress.	PostgreSQL
6	Cloud Database	Managed database service for persistent cloud storage.	PostgreSQL / Supabase
7	File Storage	Stores uploaded study materials and resource files.	Cloud Storage
8	External API-1	Email/notification service for reminders and account messages.	Email API / SMTP
9	External API-2	Optional authentication or calendar integration.	Google Calendar API
10	Machine Learning Model	Optional AI assistance for study recommendations and resource summarization.	AI/LLM API
11	Infrastructure (Server / Cloud)	Deploys the Study Buddy API and database services.	Cloud Server / Docker

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used.	FastAPI, Pydantic, SQLAlchemy, Uvicorn
2	Security	Authentication, password protection, API access control and secure communication.	JWT, HTTPS, password hashing, CORS
3	Scalable Architecture	API designed as a modular service with separate API, business logic and database layers.	3-Tier Architecture, REST API, Docker
4	Availability	Application can be deployed on reliable cloud infrastructure with managed database services.	Cloud deployment, health checks, managed PostgreSQL
5	Performance	Efficient API queries, pagination and caching can reduce response time and database load.	Async FastAPI, database indexing, caching

References:

- C4 Model – software architecture diagrams
- IBM Cloud Architecture – architecture reference
- AWS Architecture – cloud architecture reference